Ex 9.4 Q 48

$$\sum_{n=1}^{\infty} \frac{\operatorname{arcsec} n}{n^{1.3}}$$

$x \geq 0 \Rightarrow 0 \leq \operatorname{arcsec} n < \frac{\pi}{2}$

NOT Equal

Note that $0 \leq \frac{\operatorname{arcsec} n}{n^{1.3}} < \underbrace{\frac{\pi/2}{n^{1.3}}}$ for $n \geq 1$

Important to use DCT!

$\sum \frac{\pi/2}{n^{1.3}}$ conv. $(1.3 > 1)$

$\therefore \sum \frac{\operatorname{arcsec} n}{n^{1.3}}$ conv.

by DCT.

$$\sinh n = \frac{e^n - e^{-n}}{2}$$

$$\cosh n = \frac{e^n + e^{-n}}{2}$$

$$\tanh n = \frac{\sinh n}{\cosh n} = \frac{e^n - e^{-n}}{e^n + e^{-n}} = \frac{1 - e^{-2n}}{1 + e^{-2n}} \to 1$$

$$\coth n = \frac{1}{\tanh n} \to \frac{1}{1} = 1$$

$$n^{\frac{1}{n}} = \sqrt[n]{n} \to 1$$

$$\operatorname{sech} n = \frac{1}{\cancel{\cot}\cosh n} = \frac{2}{e^n + e^{-n}} \to 0$$

Q50 $\sum \frac{\tanh n}{n^2}$ $\frac{\tanh n}{n^2} \approx \frac{1}{n^2}$

LCT: $a_n = \frac{\tanh n}{n^2}$, $b_n = \frac{1}{n^2}$

$\lim \frac{a_n}{b_n} = \cdots = 1 > 0$

$\therefore \sum \frac{\tanh n}{n^2}$ conv. by LCT

($\because \sum \frac{1}{n^2}$ conv. by p-series test)

Q51 $\sum \frac{1}{n \sqrt[n]{n}}$ (LCT with $\frac{1}{n}$)

$$\sum \frac{1}{n}\underbrace{\left(1+\frac{2026}{n}\right)^n}_{e^{2026}} \approx \frac{1}{n} \qquad \left(1+\frac{x}{n}\right)^n \to e^x$$

Idea: LCT with $\frac{1}{n}$

$$\left.\begin{array}{l}\{a_n\} \text{ div.}\\ \{b_n\} \text{ conv.}\end{array}\right\} \{a_n+b_n\} \text{ div.}$$

<u>Pf.</u> (Contradiction) Suppose $\{a_n+b_n\}$ conv.

$$a_n = \underbrace{a_n+b_n}_{\text{conv.}} - \underbrace{b_n}_{\text{conv}} \Rightarrow \{a_n\} \text{ conv.}$$

(by Limit Laws)

Ex. 9.5 Q29

$$\sum\left(\frac{1}{n}-\frac{1}{n^2}\right)$$

① $\frac{1}{n}-\frac{1}{n^2}=\frac{n-1}{n^2} \sim \frac{n}{n^2}=\underline{\underline{\frac{1}{n}}}$ LCT

② $\underbrace{\sum\left(\frac{1}{n}-\frac{1}{n^2}\right)}_{\text{div.}} = \underbrace{\sum\frac{1}{n}}_{\text{div.}} - \underbrace{\sum\frac{1}{n^2}}_{\text{conv.}}$

Ex 9.5 Q 23

$$\sum \frac{2+(-1)^n}{(1.25)^n} =: \sum a_n$$

① (Geom) $\dfrac{2+(-1)^n}{(1.25)^n}$

$$= \underbrace{\frac{2}{(1.25)^n}}_{\text{Conv.}} + \underbrace{\left(-\frac{1}{1.25}\right)^n}_{\text{Conv.}}$$

② (Root)

$$\sqrt[n]{|a_n|}$$

$$= \frac{\sqrt[n]{2+(-1)^n}}{1.25}$$

(Sq. Thm)

$$1 \leq 2+(-1)^n \leq 3$$

Div $\longrightarrow$ Dont have limit

$\downarrow$ $\quad \forall l \in \mathbb{R},\ a_n \not\to l.$

∞ or $-\infty$

Ratio Test

$$\lim \left| \frac{a_{n+1}}{a_n} \right| = L$$

$L > 1$, ~~$\cdot$~~ $0 < L < 1$, $L = \infty$

Inconclusive: $L = 1$, L does not exists

$(L \neq \infty)$

$\left| \frac{a_{n+1}}{a_n} \right|$ is div.

$$\lim |a_n| \neq 0 \Rightarrow \lim a_n \neq 0$$

Contrapositive: $\lim a_n = 0 \Rightarrow \lim |a_n| = 0$

(Use def.) ε-N Language

$$\underbrace{|a_n|}_{=\,||a_n||} < \varepsilon \text{ for all } n \geq N$$

Ex 9.6 Q42

$$\sum (-1)^n \sqrt{n^2+n} - n$$

$$a_n = (-1)^n \sqrt{n^2+n} - n$$

$$|a_n| = \sqrt{n^2+n} - n$$

$$= \frac{n^2+n-n^2}{\sqrt{n^2+n}+n} = \frac{n}{\sqrt{n^2+n}+n}$$

$$= \frac{1}{\sqrt{1+\frac{1}{n}}+1} \to \frac{1}{2} \neq 0$$

$$\lim |a_n| \neq 0 \Rightarrow \lim a_n \neq 0$$

$\therefore \sum a_n$ div. by n-term test

$a_n \to \infty$

$\forall$ $M \in \mathbb{R}$, $\exists N \in \mathbb{N}$ s.t. $a_n > M$ $\forall n \geq N$

$M > 0$

$$a_n = n + (-1)^n$$

$$0, 3, 2, 5, 4, 7, \ldots$$

Ex. 9.6 Q21

$$\sum (-1)^n \frac{1}{n+3}$$

$\Downarrow$

Conv. Cond.

(AST) $b_n = \dfrac{1}{n+3}$

① $b_n \geq 0$ (Clear)

② b_n is decreasing

③ $b_n \to 0$ (Easy)

Since $n+3$ is inc., $\dfrac{1}{n+3}$ is dec.